
TECHNICAL SPECIFICATIONS
Little Goose Protective Relay Test Set

1.0 INTENT

The following specifications pertain to the purchase of a complete protective relay test set and specified accessories.

2.0 SCOPE

The successful offeror must provide equipment for the specifications listed in section 3.0 entitled SPECIFICATIONS. All equipment must be in new, unused condition in sealed original packaging from the manufacturer, and manufactured within the last 12 months of contract award. The equipment must meet or exceed the characteristics mentioned in the SPECIFICATIONS.

3.0 EQUIPMENT

The successful offeror must provide the following quantities of listed equipment

Item	Description	Qty	U/M
1	Complete Relay Test Set	1	ea
2	Complete Relay Test Set Expansion/Amplifier Module	1	ea
3	Complete Full Edition of Relay Test Set Software	1	ea
4	Accessory Package	1	ea
5	Test Set Training Session	1	ea

4.0 EQUIPMENT SPECIFICATIONS

The following are specifications for the equipment mentioned in Section 3.0. The equipment must be supplied to meet the following specifications:

Item 1: Complete Relay Test Set

- Internal Power Supply:
 - 100-240VAC input @ 50/60Hz with standard IEC socket
- Operating/storage environment conditions:
 - Ambient operating temperature: +32F to +122F
 - Storage temperature: -13F to +158F
 - Ambient operating humidity: 5-95% relative
- Computer Communication Capabilities:
 - (1) Ethernet communication port
 - (1) USB type B port
- Relay test set must be UL Listed
- Relay test set must be FCC compliant
- Test set must weigh less than 50lbs
- AC Voltage Outputs:
 - (1) three phase output line to neutral, output adjustable from 0 to 300VAC RMS, 300VA total
 - (1) single phase output line to neutral, output adjustable from 0-600VAC RMS, 300VA
 - All outputs capable of outputting sinusoidal signals ranging 10 to 3000Hz with a 0.001Hz or better resolution
 - Frequencies must also be adjustable by phase, with 0.001 degrees of resolution or better
 - Accuracy at full voltage: less than 0.05% error
 - Total harmonic distortion less than 0.05%
 - Resolution: 5mV for 3 phase output, 10mV for single phase output
 - Rated for continuous output at maximum voltage and current rating.
- DC Voltage Output
 - (1) DC voltage output adjustable from 0-250VDC at 0.2A, 0-132VDC at 0.4A, 0-66VDC at 0.8A
 - Rated for continuous output at maximum voltage and current
- AC Current Outputs
 - These outputs act as a constant current source
 - (2) sets of AC current outputs, 3 phase line to neutral
 - Capable of being paralleled with each other and external expansion/amplifier module
 - Capable of operating in 6 phase configuration
 - With all 6 phases operating, capable of supplying 32 amps per phase
 - With 3 phases operating, capable of supplying 64 amps per phase
 - With single phase output, capable of supplying 128 amps
 - Maximum open circuit voltage of 140V peak between two phases
 - All outputs capable of outputting sinusoidal signals ranging 10 to 1000Hz with a 0.001Hz resolution
 - Frequencies must also be adjustable by phase, with 0.001 degrees of resolution
 - Accuracy at full current: less than 0.05% error
 - Total harmonic distortion less than 0.05%
 - Resolution: 1mA
 - Rated for continuous output at maximum current rating.
- Digital outputs
 - Provided with (4) solid state controlled outputs
 - Less than 0.1ms pickup/dropout time
 - 0.5A @ 250VAC make/break capability
 - 600VDC isolation
 - Provided with (4) relay-controlled outputs

- Rated 8A at 300VAC/VDC make/break capability
 - Less than 10ms pickup/dropout time
- Low Level Outputs
 - Provided with (12) low level output sources
 - Capable of outputting DC and AC signals ranging from 1Hz to 3kHz
 - Peak voltage output is 10V
- Analog/Binary inputs
 - Provided with 10 inputs
 - Each input must be programmable to receive and measure analog/AC input signals as well as monitor a binary input signal
 - Each input must be scalable to measure 0-100mV, 1V, 10V 100V, and 600V
 - Inputs must be 600V CAT IV rated
 - Sample rate must be 10kHz for each input, 100us resolution
 - Each input must have an adjustable trigger level in software
 - Input impedance: 150kΩ
- Analog DC inputs
 - Separate inputs from analog/binary inputs mentioned before
 - (1) input for measuring a $\pm 20\text{mA}$ DC instrumentation signal
 - (1) input for measuring a $\pm 10\text{VDC}$ instrumentation signal
- Time synchronization
 - Internal time clock must have a frequency drift less than 0.37ppm/24hrs
 - Timing accuracy of <5us with external devices
 - Capable of hosting a PTP time server
 - Provided with an RF level input for an external GPS antenna: SMA, N-type, TNC, or BNC; or provided with an adapter to fit these types
 - Capable of receiving and transmitting IRIG-B signals via a ST multimode fiber-optic female port or supplied adapter
- The test set must be supplied with the following test leads. All test leads must be 1000V cat III/600V Cat IV rated dual insulation with inner layer being a contrasting color to provide indication if there is damage or wear on the lead assembly and plugs/jacks must be safety rated:
 - (6) black and (6) red, 6 foot test leads with 4mm safety banana jack plugs, male and female end
 - (12) flexible terminal block to 4mm jack adapters, black, male
 - (4) black flexible male 4mm jumper wires, 2.4in
 - (6) black and (6) red test lead adapters for test leads mentioned above with retractable lead sleeve, 4mm female
 - (1) grounding cable with clamp and cable lug, green, 20 foot length
 - Soft bag for test leads
 - (12) narrow terminal block to 4mm jack adapters, black, male
 - (12) test lead alligator clips, insulated, 4mm male
 - (1) combination cable to interface with combination socket on relay tester if tester is provided with socket, black, blue, red, yellow wires
- Test set must be provided with 1 heavy duty transport case, approved by the relay test set manufacturer
 - Wheels and extendable handle
 - Provide protection against dust, water, and mechanical damage
 - Must be able to contain all test leads mentioned above with relay test set
 - Contains locking latches
- Relay test set must be fully controllable and programmable via supplied programming and interface software
- All inputs and outputs must have universal 4mm safety-rated banana jacks, 1000V CAT III/600V CAT IV
- The relay test set must incorporate a built-in grounding stud to bond it to the system ground with a ground lead
- Must be capable of recording all digital and analog input/output sources simultaneously

Item 2: Relay Test Set Expansion/Amplifier Module

- Internal Power Supply:
 - 100-240VAC input @ 50/60Hz with standard IEC socket
- Operating/storage environment conditions:
 - Ambient operating temperature: +32F to +122F
 - Storage temperature: -13F to +158F
 - Ambient operating humidity: 5-95% relative
- Computer Communication Capabilities:
 - (1) Ethernet communication port
 - (1) USB type B port
- Must be UL Listed
- Must be FCC compliant
- Must weigh less than 50lbs
- Directly compatible with Item 1. Communicates and is controlled directly by Item 1 via low level signals and ethernet cable.
- Provides (1) additional three phase voltage output meeting the same AC voltage output specifications as Item 1
- Provides (2) additional three phase current outputs meeting the same AC current output specifications as Item 1
- Outputs must be capable of being paralleled with the outputs on Item 1 to increase output current and voltage test capability
- Must be the same manufacturer as the Relay Test set and be a specified compatible accessory
- Provided with the following test leads. Leads must be 1000V cat III/600V Cat IV rated dual insulation with inner layer being a contrasting color to provide indication if there is damage or wear on the lead assembly and plugs/jacks must be safety rated
 - (6) black and (6) red, 6 foot test leads with 4mm safety banana jack plugs, male and female end
 - (12) flexible test lead adapters with retractable lead sleeves
 - (12) flexible terminal strip adapters, black
 - (4) flexible male-male 4mm flexible jumpers, 2.4" length, black
- Must be provided with 1 heavy duty transport case, approved by the manufacturer
 - Wheels and extendable handle
 - Provide protection against dust, water, and mechanical damage
 - Must be able to contain all test leads mentioned above and expansion module
 - Contains locking latches

Item 3: Relay Test Software

- Provide programming and interface software for the relay test set
- The software must be:
 - Provided directly from the test set manufacturer
 - Provided in full either on a USB flash drive or external USB Hard Drive
 - Provided with a perpetual license, transferrable between computers without the need for internet access
 - Able to interface with all input and output sources of the relay test set
 - Capable of recording input and output sources at the mentioned resolution, exportable to a *.csv file type for record keeping
 - Be the complete, full, professional version of the software package available with the relay test set
- The Software test features must include:
 - Automatic test plans (including configuration steps) for common protective relays, specifically Schweitzer Engineering and General Electric protective relays, both electromechanical and electronic
 - The ability to perform manual tests on relays
 - Sequence logger between test set's outputs and inputs for determining attached relay's operating times
 - Oscilloscope for displaying operating times with playback feature. These times must be able to be saved in a format to be opened in the future, as well as being exported to a *.csv file type
 - Ability to generate output waveforms with imposed harmonics
 - Ability to compensate for circuit breaker delay timing
 - Have ramp and pulse ramp output and measuring capabilities
 - Test positive, negative, and zero sequence overcurrent characteristics
 - Simulate line distance faults for distance relaying
 - Simulate VI curves for distance relays
 - Test reclosure functions on relays
 - Advanced differential protection characteristics and test inrush blocking on relays
 - Provided with a means for testing and verifying annunciation on attached relays
 - Advanced power and power quality tests and visualization for power quality and revenue relays
 - Ability to simulate ground faults and transients on attached relays
 - Ability to verify synchronization abilities of generator protection/synchrocheck relays
 - Ability to test and verify accuracy of multifunction energy meters
 - Ability to simulate power quality events
 - Fully programmable custom simulation events
 - Ability to take analog measurements and transient records
 - Full ability to update the software and firmware on the relay test set

Item 4: Accessories

- Interface Cables
 - (2) 10' ethernet patch cable
 - (2) 10' USB connection cable, type B to type A
 - If item 1 or item 2 are provided with a DB9 serial port, provide a serial to USB converter
- Supplied with (2) removable 10 foot power cords compatible with item 1 and item 2 for plugging into a standard 125VAC NEMA 5-15R receptacle
- (3) Current Clamp Probes
 - Must be an active-style current clamp with 9 volt battery power supply
 - Must be capable of measuring DC current and AC RMS current waveforms up to 10kHz
 - Must have positive and negative universal banana leads to interface with test set
 - Current measurement range 5mA to 80A with selector switch to select scale. Output must be in levels of millivolts per measured amps
 - Capable of clamping onto cable ~0.5inches in diameter
- CT Polarity checker
 - Injects signal into the primary of a CT, synced with the relay test set to verify correct polarity of a current transformer
 - Battery powered by 1.5V AA's

Item 5: Training Session

- Provide (1) technician, an employee of the test-set manufacturer or certified directly by the manufacturer, to train personnel on the use of the relay test set provided in this contract
- Training session must be at a minimum of 8 hours in length
- Must be in person at Little Goose Lock and Dam
- Must include travel and per-diem to Little Goose Lock and Dam for the training session
- Must be coordinated in advance with Little Goose personnel
- Must occur within 12 months of the delivery of the relay test set and not before delivery
- Must cover the following:
 - Initial setup and configuration of the relay test set
 - Overview of operation of the relay test set
 - Overview of the relay test set software
 - Overview of each test the relay test set can perform
 - General maintenance procedures on the relay test set
- Must provide training materials including:
 - Informational booklets/Cheat Sheets
 - Devices to demonstrate tests performed by the relay test set

5.0 WARRANTY

The above-mentioned equipment must be covered by a standard manufacturer warranty not less than 12 months in duration from date of delivery with no cost to the Government.

6.0 EQUIPMENT INFORMATION

Prior to contract acceptance, the successful offeror must supply, for Government approval, manufacturer's product datasheets for all sizes and types of materials covered under this contract. This information must include factory test results on the equipment as well as calibration certifications from a third party.

7.0 OPERATION INSTRUCTIONS

The successful offeror must supply a physical copy of the manufacturer's operation instructions for the relay test set and software. These instructions must also contain recommended maintenance procedures for the relay test set.

8.0 SHIPMENT AND DELIVERY

All items must be delivered FOB Destination to Little Goose Dam within 25 weeks of award. The successful offeror must be responsible for packing and shipping the components in a manner that will ensure undamaged and complete arrival to Little Goose Dam. No shipments are allowed other than normal business hours of Monday-Thursday 7:00 AM-3:00PM, excluding holidays.

All shipments and invoices must be delivered to:

Little Goose Dam
Attn: Little Goose Tech Section
1001 Little Goose Dam Road
Dayton, WA 99328
Ph. (509) 404-3259 / (509) 404-3279
